Foundation of Programming

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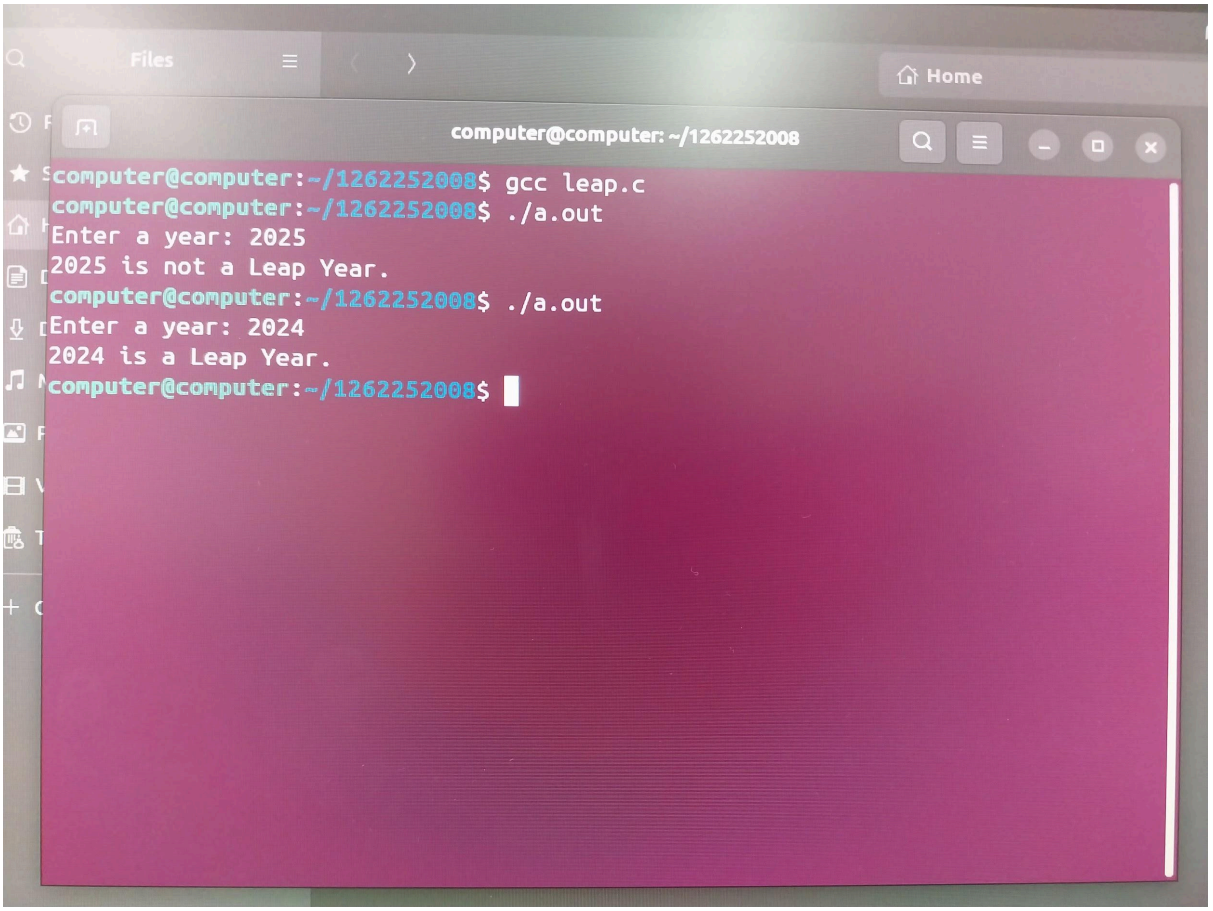
Division: FY BTech CSE CsF Div 13

Assignment 1:

Leap year

```
#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("%d is a Leap Year.\n", year);
    } else {
        printf("%d is not a Leap Year.\n", year);
    }

    return 0;
}
```



Assignment 2:

Menu driven Operations

```
#include <stdio.h>

int main() {
    int choice;
    float num1, num2;
```

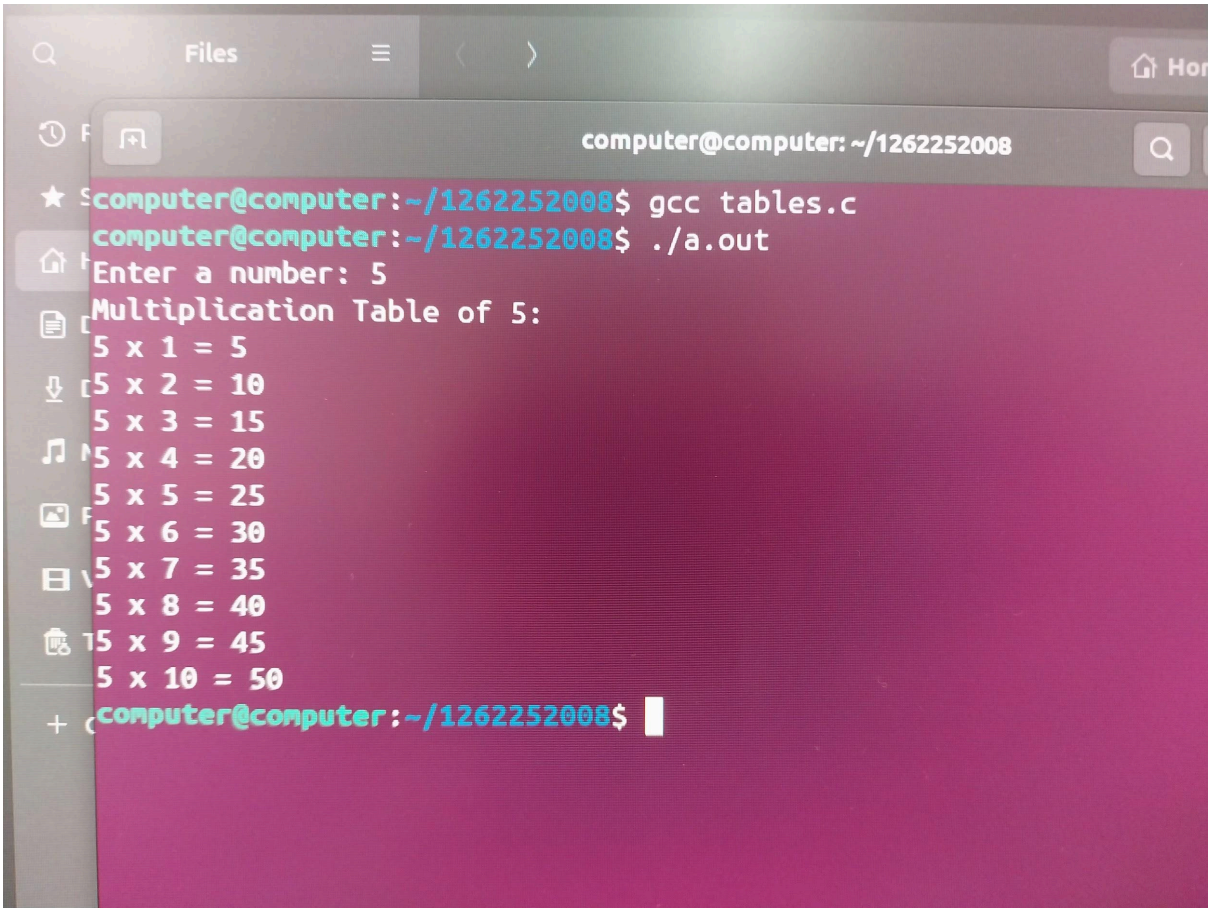
```
    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    printf("Enter two numbers: ");
    scanf("%f %f", &num1, &num2);

    switch(choice) {
        case 1:
            printf("Result = %.2f\n", num1 + num2);
            break;

        case 2:
            printf("Result = %.2f\n", num1 - num2);
            break;

        case 3:
            printf("Result = %.2f\n", num1 * num2);
            break;

        case 4:
            if (num2 != 0)
                printf("Result = %.2f\n", num1 / num2);
            else
                printf("Error!\n");
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

```
computer@computer:~/1262252008$ gcc ops.c
computer@computer:~/1262252008$ ./a.out
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Enter two numbers: 2 3
Result = 5.00
computer@computer:~/1262252008$ ./a.out
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 2
Enter two numbers: 2 3
Result = -1.00
computer@computer:~/1262252008$ ./a.out
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 3
Enter two numbers: 2 3
Result = 6.00
computer@computer:~/1262252008$ ./a.out
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 4
Enter two numbers: 2 3
Result = 0.67
computer@computer:~/1262252008$
```

Assignment 3:

Multiplication Tables

```
#include <stdio.h>

int main() {
    int num, i;

    printf("Enter a number: ");
    scanf("%d", &num);

    printf("Multiplication Table of %d:\n", num);

    for(i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", num, i, num * i);
    }

    return 0;
}
```

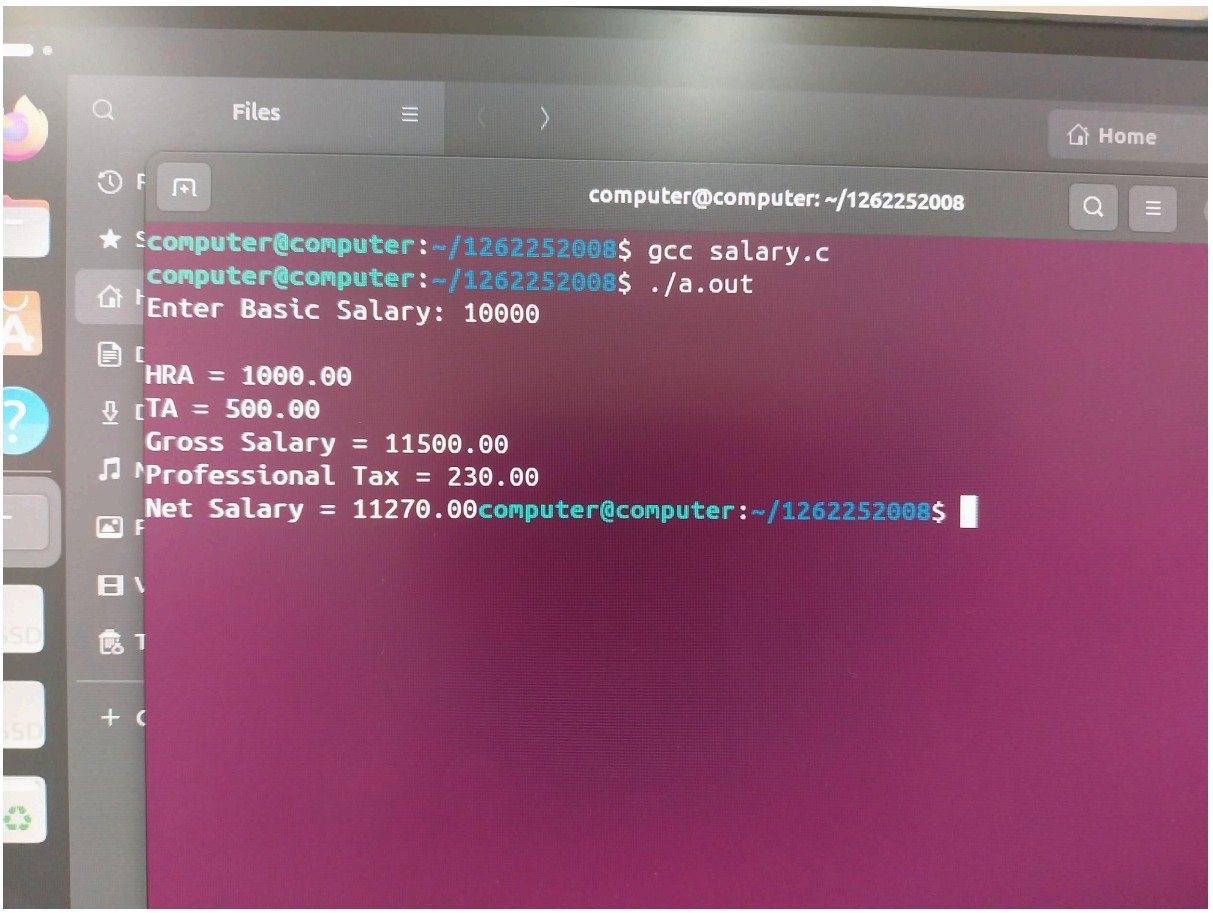


Assignment 4:

Employee Salary

```
#include <stdio.h>

int main() {
    float basic, hra, ta, gross, tax, net;

    printf("Enter Basic Salary: ");
    scanf("%f", &basic);

    hra = 0.10 * basic;
    ta = 0.05 * basic;
    gross = basic + hra + ta;
    tax = 0.02 * gross;
    net = gross - tax;

    printf("\nHRA = %.2f", hra);
    printf("\nTA = %.2f", ta);
    printf("\nGross Salary = %.2f", gross);
    printf("\nProfessional Tax = %.2f", tax);
```

```
    printf("\nNet Salary = %.2f", net);

    return 0;
}
```

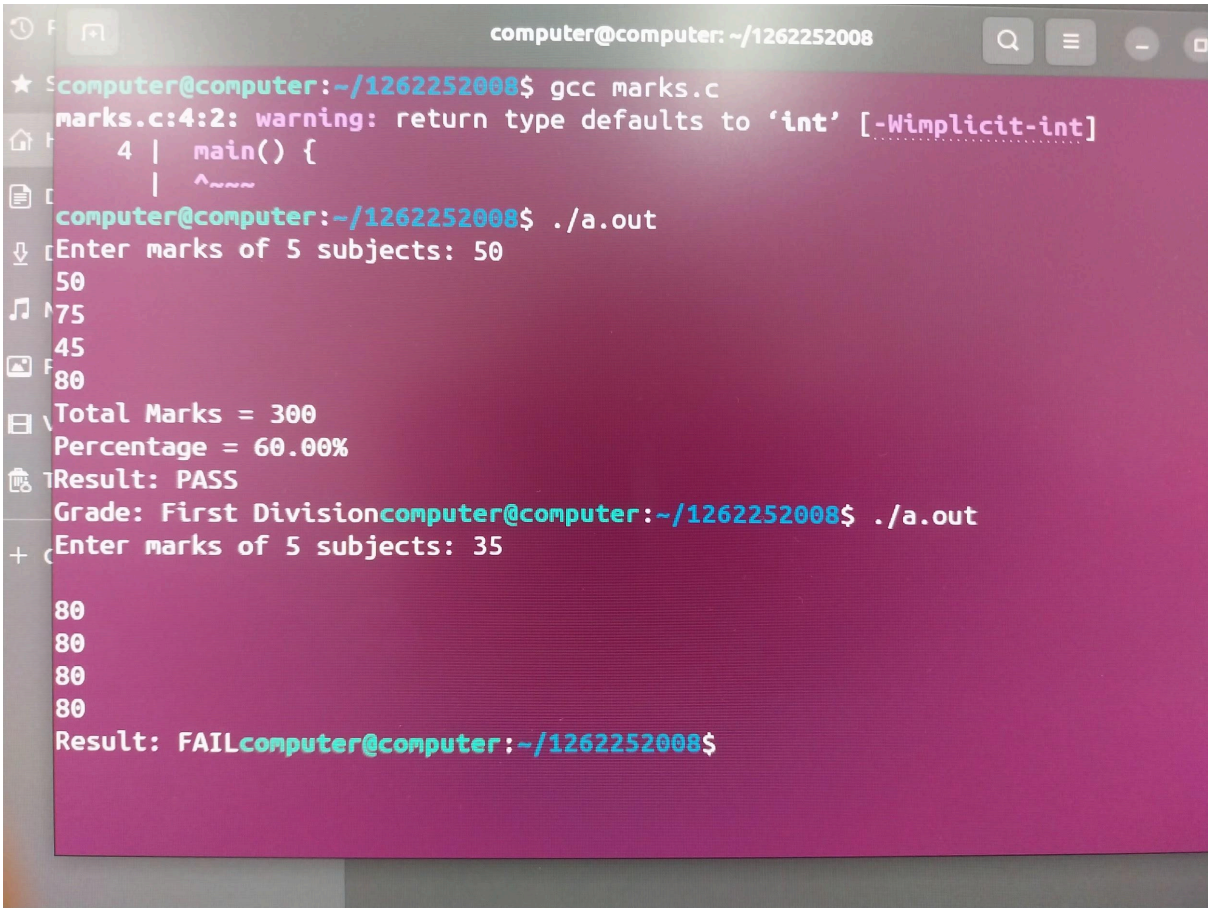

Files
Home
computer@computer: ~/1262252008
computer@computer:~/1262252008$ gcc salary.c
computer@computer:~/1262252008$ ./a.out
Enter Basic Salary: 10000
HRA = 1000.00
TA = 500.00
Gross Salary = 11500.00
Professional Tax = 230.00
Net Salary = 11270.00computer@computer:~/1262252008$

Assignment 8:

Percentage of Marks of student.

```
#include <stdio.h>

 main() {
   int m1, m2, m3, m4, m5, total;
   float percentage;

   printf ("Enter marks of 5 subjects: ");
   scanf ("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

   if (m1 < 40 || m2 < 40 || m3 < 40 || m4 < 40 || m5 < 40) {
       printf ("Result: FAIL");
   } else {
       total = m1 + m2 + m3 + m4 + m5;
       percentage = total / 5.0;

       printf ("Total Marks = %d\n", total);
       printf ("Percentage = %.2f%%\n", percentage);
       printf ("Result: PASS\n");

       if (percentage >= 75)
           printf("Grade: Distinction");
```

```
        else if (percentage >= 60)
            printf("Grade: First Division");
        else if (percentage >= 50)
            printf("Grade: Second Division");
        else
            printf("Grade: Third Division");
    }

    return 0;
}
```

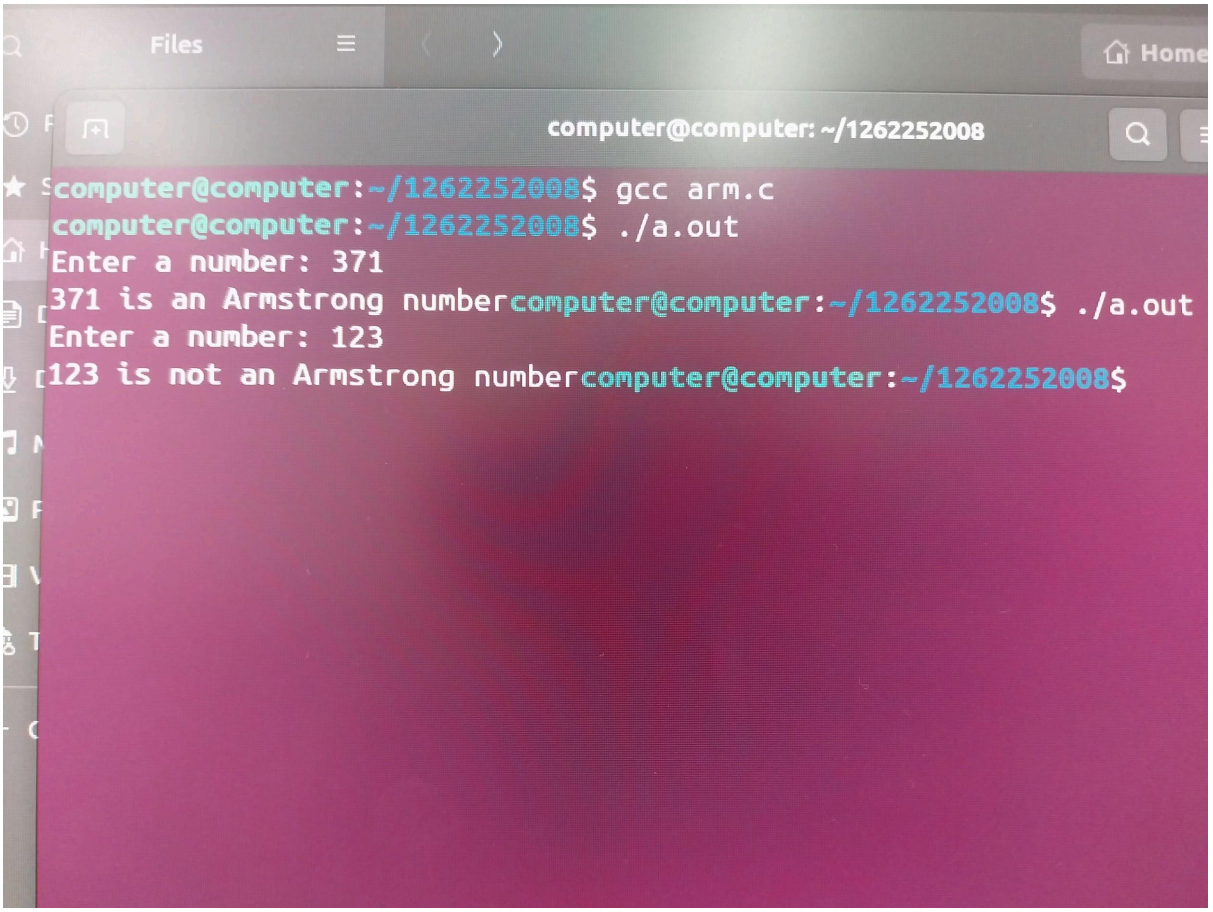


Assignment 9:

Armstrong Number

```
#include <stdio.h>

int main() {
```

```
    int num, temp, digit, sum = 0;

    printf("Enter a number: ");
    scanf("%d", &num);

    temp = num;

    while (temp > 0) {
        digit = temp % 10;
        sum += digit * digit * digit;
        temp = temp / 10;
    }

    if (sum == num)
        printf("%d is an Armstrong number", num);
    else
        printf("%d is not an Armstrong number", num);

    return 0;
}
```

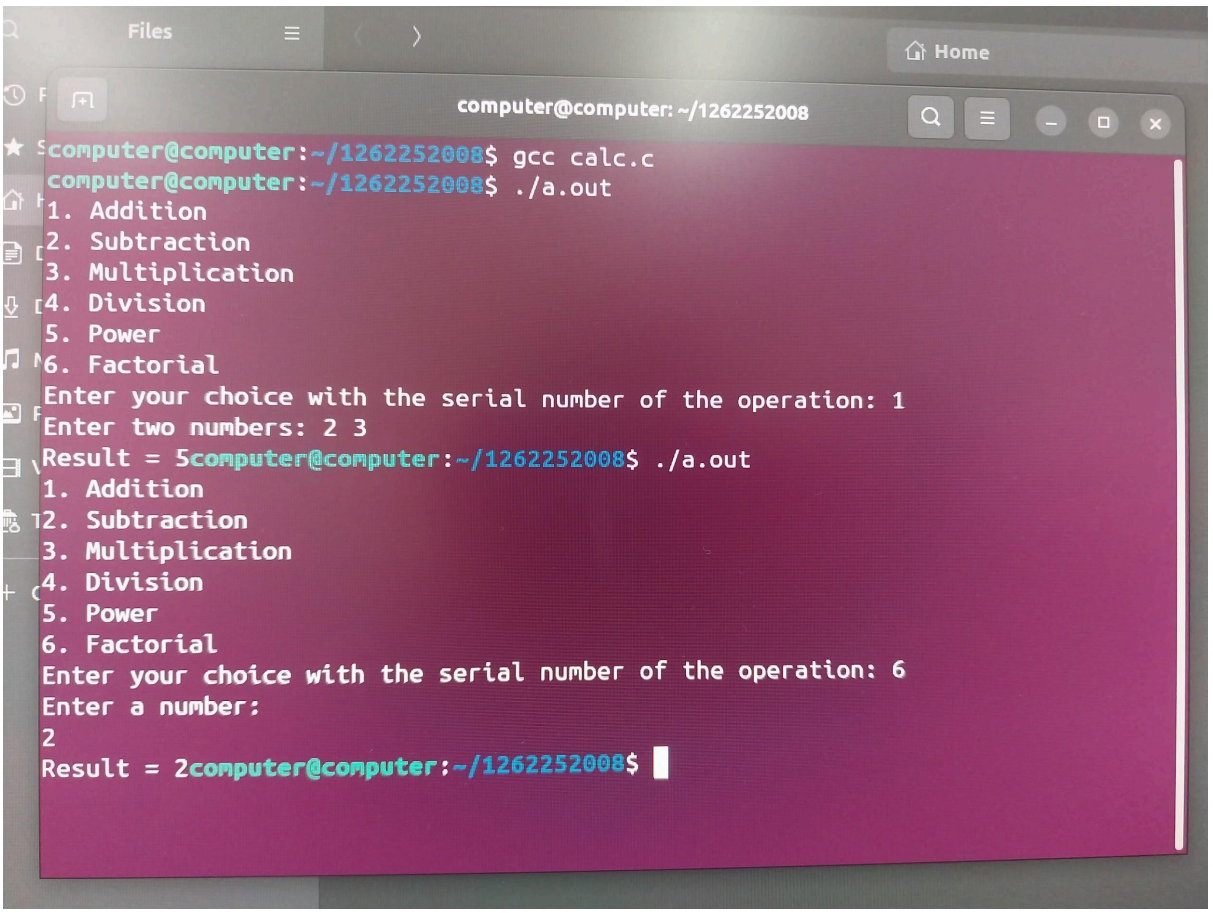


Assignment 10:

Simple Calculator (without switch case)

```
#include <stdio.h>

int main() {
    int choice, a, b, i;
    int result = 1;

    printf("1. Addition\n2. Subtraction\n3. Multiplication\n4. Division\n5. Power\n6.
Factorial\n");
    printf("Enter your choice with the serial number of the operation: ");
    scanf("%d", &choice);

    if (choice >= 1 && choice <= 4) {
        printf("Enter two numbers: ");
        scanf("%d %d", &a, &b);
    } else {
        printf("Enter a number: ");
        scanf("%d", &a);
    }

    if (choice == 1)
        printf("Result = %d", a + b);

    else if (choice == 2)
        printf("Result = %d", a - b);

    else if (choice == 3)
        printf("Result = %d", a * b);

    else if (choice == 4) {
```

```
        if (b != 0)
            printf("Result = %.2f", (float)a / b);
        else
            printf("Error");
    }

    else if (choice == 5) {
        printf("Enter power: ");
        scanf("%d", &b);
        result = 1;
        for (i = 1; i <= b; i++)
            result *= a;
        printf("Result = %d", result);
    }

    else if (choice == 6) {
        result = 1;
        for (i = 1; i <= a; i++)
            result *= i;
        printf("Result = %d", result);
    }

    else
        printf("Invalid choice");

    return 0;
}
```


Files
Home
computer@computer: ~/1262252008
computer@computer:~/1262252008$ gcc calc.c
computer@computer:~/1262252008$ ./a.out
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power
6. Factorial
Enter your choice with the serial number of the operation: 1
Enter two numbers: 2 3
Result = 5computer@computer:~/1262252008$ ./a.out
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power
6. Factorial
Enter your choice with the serial number of the operation: 6
Enter a number:
2
Result = 2computer@computer:~/1262252008$

Assignment 11:

Operations

```
#include <stdio.h>
#include <math.h>

int main() {
    int n, i, flag = 1;

    scanf("%d", &n);

    printf("Square: %d\n", n*n);
    printf("Cube: %d\n", n*n*n);
    printf("Square root: %.2f\n", sqrt(n));

    for(i=2;i<n;i++)
        if(n%i==0) flag=0;

    printf("Prime: %s\n", flag?"Yes":"No");

    return 0;
}
```

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myCompiler

English

Enter a title...

C

```c
#include <stdio.h>
#include <math.h>

int main() {
    int n, i, flag = 1;

    scanf("%d", &n);

    printf("Square: %d\n", n*n);
    printf("Cube: %d\n", n*n*n);
    printf("Square root: %.2f\n", sqrt(n));

    for(i=2;i<n;i++)
        if(n%i==0) flag=0;

    printf("Prime: %s\n", flag?"Yes":"No");

    return 0;
}
```

5

Output

```
Square: 25
Cube: 125
Square root: 2.24
Prime: Yes

[Execution complete with exit code 0]
```

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